

Comparative Evaluation of AI Agents

Browser-Only Word to PDF Converter

Detailed Assignment Report

Task	Same browser-only DOCX to PDF converter given to every agent
Review date	17 September 2026
Primary focus	Requirement fit, privacy communication, UX, visible features and deployment quality
Security note	Public-output evaluation; no-upload claims are not treated as cryptographic proof

1. Objective and Common Task

The purpose of this assignment is to compare how different AI app-building and coding agents respond when they receive the same software requirement. Using one identical task makes the comparison more meaningful because differences in the final output come from each agent's interpretation, design choices, implementation style and deployment workflow rather than from different prompts.

Common task: Build a minimal Word (.docx) to PDF converter. The user uploads a Word file and receives a PDF. Conversion must happen entirely in the browser, with no document upload to a server. The interface should be simple, responsive and clearly communicate privacy.

2. Evaluation Method

Each live output was inspected using the same five categories. Scores are comparative and based on the publicly observable result.

- Task fit - how closely the output matches the requested browser-only converter.
- UI / UX - clarity, simplicity and how easy the main action is to understand.
- Privacy message - how clearly the page explains local processing and no upload.
- Visible features - upload guidance, limits, status information and task-focused completeness.
- Deployment polish - whether the result looks like a finished, accessible application rather than only a development preview.

Verification limitation: the browsing channel could inspect live pages and public text, but it could not execute every file upload/conversion workflow. Therefore the report does not claim that any local-only security statement has been independently proven. Source-code review or browser network tracing would be required for that.

3. Live Outputs Reviewed

Agent / Platform	Host	Live output	Score
Emergent	Emergent preview	pdf-maker-274.preview.emergentagent.com/	76/100
Lovable	Lovable	local-doc-pdf.lovable.app/	91/100
Replit Agent	Replit	private-word-to-pdf--laplot8.replit.app/	75/100
Bolt	Bolt	client-side-word-to-dmue.bolt.host/	86/100
Vercel-hosted build	Vercel	word-to-pdf-zeta.vercel.app/	93/100
Zite	Zite	opgyzvetxq.zite.so/	76/100
Base44	Base44	local-pdf-flow.base44.app/	84/100
OpenCode	Vercel deployment	word-to-pdf-yesb-eight.vercel.app/	90/100
Google Antigravity	Vercel deployment	word-to-pdf-converter-gray.vercel.app/	93/100

4. Score Comparison

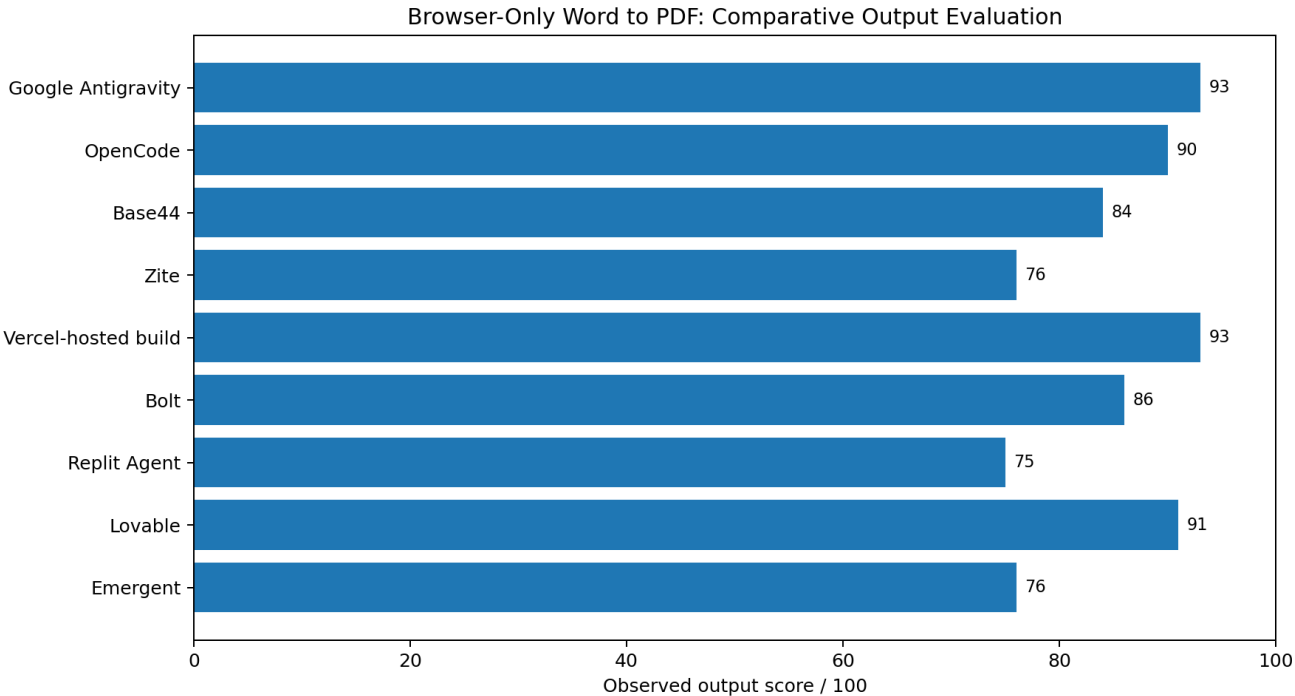


Figure 1. Overall comparative output score based on the five-category rubric.

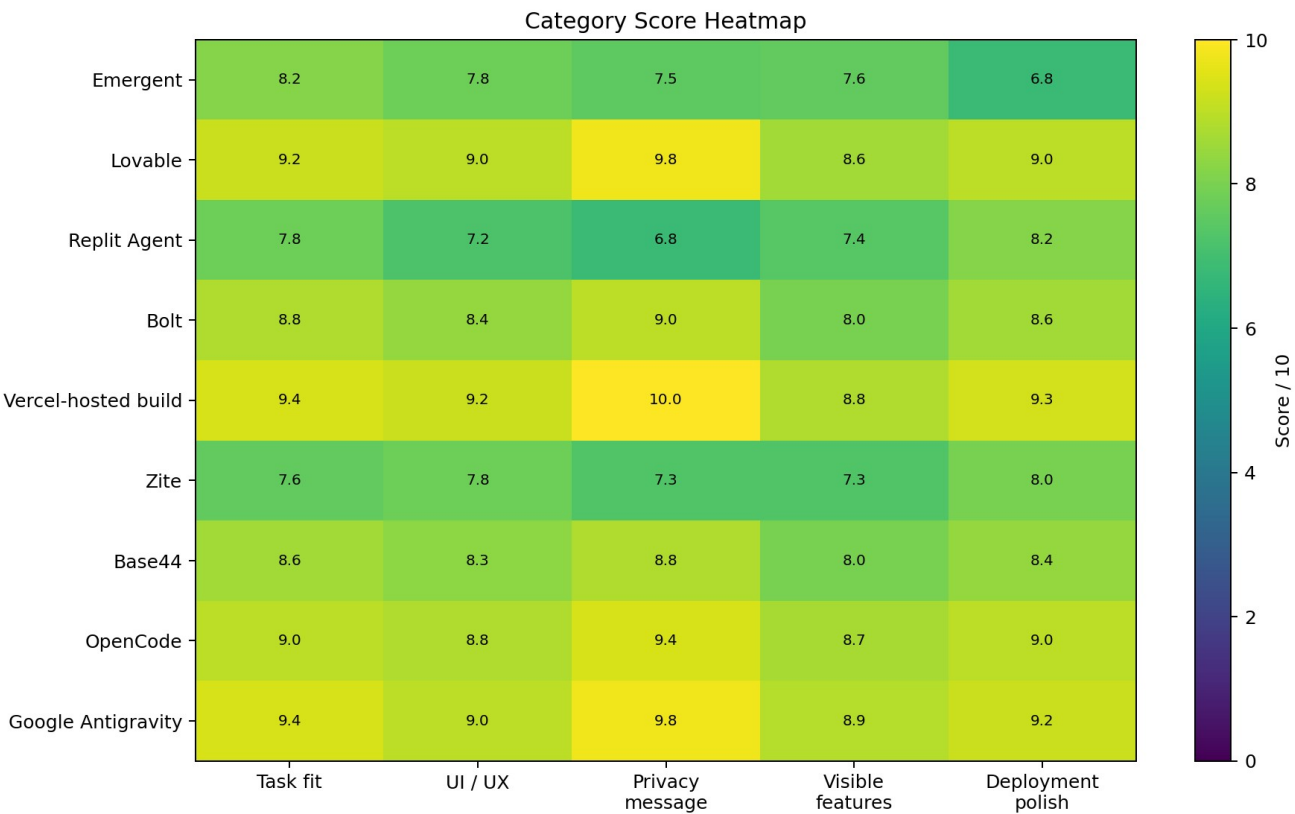


Figure 2. Category-level comparison. Higher values indicate stronger observable alignment with the assignment.

5. Detailed Agent-by-Agent Evaluation

5.1 Emergent - 76/100

Platform / agent properties: Agentic prompt-to-production platform for full-stack web/mobile apps, with preview, testing, deployment and integrations.

Observed output: Live link resolved through an Emergent iframe/preview surface. The use of a preview URL makes it look more like a development result than a final production deployment.

Strongest feature: Fast end-to-end app generation with a working hosted preview.

Main limitation: Deployment maturity is weaker than the dedicated production URLs in this comparison, and privacy-specific interface text was not fully extractable during inspection.

Task fit: 8.2/10 | UI / UX: 7.8/10 | Privacy message: 7.5/10 | Visible features: 7.6/10 | Deployment polish: 6.8/10

5.2 Lovable - 91/100

Platform / agent properties: Conversational app and website builder with live iteration, visual editing, publishing and managed hosting.

Observed output: The page clearly states that conversion happens in the browser, files never leave the device, and DOCX files up to 25 MB are accepted. The upload action is immediately understandable.

Strongest feature: Excellent privacy communication combined with a very clear drag-and-drop upload flow.

Main limitation: The visible interface is intentionally minimal, so advanced document controls are not part of the public surface.

Task fit: 9.2/10 | UI / UX: 9.0/10 | Privacy message: 9.8/10 | Visible features: 8.6/10 | Deployment polish: 9.0/10

5.3 Replit Agent - 75/100

Platform / agent properties: General software creation agent that can generate code, install dependencies, run applications and deploy them from the same environment.

Observed output: The deployed page is reachable and titled "Word to PDF". Compared with the strongest outputs, less requirement-specific information was visible to the text inspection layer.

Strongest feature: Strong developer environment and straightforward path from generated code to a live app.

Main limitation: The public-facing output communicates fewer privacy and task details on the surface available to inspection.

Task fit: 7.8/10 | UI / UX: 7.2/10 | Privacy message: 6.8/10 | Visible features: 7.4/10 | Deployment polish: 8.2/10

5.4 Bolt - 86/100

Platform / agent properties: Prompt-driven web app builder focused on fast code generation, live preview, responsive interfaces and rapid publishing.

Observed output: The live page title explicitly describes a "Private Browser-Based Converter", which closely matches the assignment requirement for client-side conversion.

Strongest feature: Strong requirement alignment and fast, focused frontend output.

Main limitation: The page content was JavaScript-heavy and not fully extractable, so deeper feature verification from the public surface was limited.

Task fit: 8.8/10 | UI/UX: 8.4/10 | Privacy message: 9.0/10 | Visible features: 8.0/10 | Deployment polish: 8.6/10

5.5 Vercel-hosted build - 93/100

Platform / agent properties: The generating agent was not specified in the provided information. Vercel is the deployment platform, so this row evaluates the output rather than attributing capabilities to an unknown agent.

Observed output: The output is branded "DocVault" and clearly states: no uploads, no servers and no tracking. It also shows DOCX support up to 30 MB and repeats that documents never leave the device.

Strongest feature: The clearest combination of privacy messaging, file-limit guidance and polished product-style presentation.

Main limitation: Agent-level properties cannot be evaluated fairly because the builder/agent name was not provided.

Task fit: 9.4/10 | UI/UX: 9.2/10 | Privacy message: 10.0/10 | Visible features: 8.8/10 | Deployment polish: 9.3/10

5.6 Zite - 76/100

Platform / agent properties: AI builder oriented toward business apps, databases, forms, workflows, websites and operational automation.

Observed output: The deployment is reachable as "Word to PDF | Zite". Zite is capable of broader business systems, but this task is a small single-purpose browser utility.

Strongest feature: Rapid deployment inside a platform that can later expand into workflows or business processes.

Main limitation: For this narrow client-side converter task, the platform is broader than necessary and the output exposes fewer task-specific details in text inspection.

Task fit: 7.6/10 | UI/UX: 7.8/10 | Privacy message: 7.3/10 | Visible features: 7.3/10 | Deployment polish: 8.0/10

5.7 Base44 - 84/100

Platform / agent properties: No-code AI builder with built-in backend, auth, storage, workflows, responsive design, testing and publishing.

Observed output: The page title explicitly identifies a private browser-based converter, showing good alignment with the requested architecture.

Strongest feature: Good prompt-to-app completeness with clear privacy-oriented positioning.

Main limitation: Base44 includes many full-stack capabilities that are unnecessary for a deliberately local-only conversion tool.

Task fit: 8.6/10 | UI / UX: 8.3/10 | Privacy message: 8.8/10 | Visible features: 8.0/10 | Deployment polish: 8.4/10

5.8 OpenCode - 90/100

Platform / agent properties: Open-source coding agent with build and read-only plan modes, provider flexibility and strong developer control through a terminal/client-server workflow.

Observed output: The deployed page title is "Word to PDF Converter | Private & Local", directly matching the local-processing requirement and presenting the result as a complete utility.

Strongest feature: Strong developer control with a clean output that keeps the privacy requirement prominent.

Main limitation: Unlike hosted no-code builders, deployment and project management remain more developer-driven.

Task fit: 9.0/10 | UI / UX: 8.8/10 | Privacy message: 9.4/10 | Visible features: 8.7/10 | Deployment polish: 9.0/10

5.9 Google Antigravity - 93/100

Platform / agent properties: Multi-step coding agent with reasoning, tools, browser control, artifacts and parallel agent workflows for end-to-end software tasks.

Observed output: The public page title is "Word to PDF | 100% Private In-Browser Converter", which is highly aligned with the assignment wording and clearly communicates the intended privacy model.

Strongest feature: Very strong requirement interpretation, privacy framing and polished deployment-oriented output.

Main limitation: The public page alone cannot prove that no file data is ever transmitted; that requires source or network-level verification.

Task fit: 9.4/10 | UI / UX: 9.0/10 | Privacy message: 9.8/10 | Visible features: 8.9/10 | Deployment polish: 9.2/10

6. Comparison Matrix

Agent	Task	UX	Privacy	Features	Deploy	Overall
Emergent	8.2	7.8	7.5	7.6	6.8	76
Lovable	9.2	9.0	9.8	8.6	9.0	91
Replit Agent	7.8	7.2	6.8	7.4	8.2	75
Bolt	8.8	8.4	9.0	8.0	8.6	86
Vercel-hosted build	9.4	9.2	10.0	8.8	9.3	93
Zite	7.6	7.8	7.3	7.3	8.0	76
Base44	8.6	8.3	8.8	8.0	8.4	84
OpenCode	9.0	8.8	9.4	8.7	9.0	90
Google Antigravity	9.4	9.0	9.8	8.9	9.2	93

7. Deployment Notes for the Provided Vercel Links

Two Vercel dashboard links were also provided. They are project-management pages rather than separate public converter outputs:

- word-to-pdf-converter project: the inspected Vercel dashboard showed "No Production Deployment" at review time. It is therefore not scored as a separate runnable application.
- word-to-pdf-yesb project: this is the Vercel project dashboard associated with the OpenCode deployment; the public word-to-pdf-yesb-eight.vercel.app URL is used in the comparison.

8. Main Findings

The most successful outputs stayed tightly focused on the actual assignment instead of adding unrelated complexity. The Vercel-hosted DocVault output and the Google Antigravity output were especially strong in privacy communication and production-style presentation. Lovable also performed very strongly because it clearly explained that conversion happens locally and displayed an explicit 25 MB DOCX limit. OpenCode produced a clear private/local result while also representing a developer-controlled workflow.

Broader full-stack builders such as Emergent, Replit, Zite and Base44 remain powerful, but their extra backend, database and workflow capabilities are not automatically useful for this task. Because the requirement specifically avoids server-side document processing, a smaller client-side implementation is often a better fit than a heavier full-stack architecture.

A key lesson from the comparison is that output quality depends not only on whether an agent can write code, but also on how well it interprets constraints. Clear privacy copy, simple upload interaction, sensible file guidance and deployment polish made the strongest results feel more complete even though the underlying task was small.

9. Conclusion

Giving the same prompt to multiple AI agents produced noticeably different levels of clarity, polish and requirement alignment. The strongest solutions did three things well: they kept the UI minimal, made local processing obvious, and presented the converter as a finished product. For future evaluations, the next step should be to add a controlled conversion test with the same DOCX file, compare the generated PDFs for formatting fidelity, and capture browser network logs to independently verify the no-upload requirement.

10. Sources and Inspection Record

- Live application links supplied for this assignment and inspected on 17 September 2026.
- Emergent official documentation and product pages: <https://help.emergent.sh/> and <https://emergent.sh/>
- Lovable official product pages: <https://lovable.dev/>
- Replit Agent official pages: <https://replit.com/products/agent>
- Bolt official app builder pages: <https://bolt.new/>
- Zite official builder site: <https://www.zite.com/>
- Base44 official features: <https://base44.com/features>
- OpenCode project documentation: <https://opencode.ai/> and its official/open-source repository documentation.
- Google Antigravity official docs: <https://www.antigravity.google/docs/agent>